WELL COMPLETION REPORT SUMMARY

Well ID: WELL-3589-NX
Operator: Northmoor Energy
Completion Date: 2024-09-03
Measured Depth (m): 3186.9
Formation: Zechstein
Completion Status: Partial

1. Summary

Well-3589-NX, operated by Northmoor Energy in the Zechstein formation, was completed on September 3, 2024, with a partial outcome. The upper zone was successfully completed, while the lower zone was isolated using a bridge plug at 2812 m MD. Completion operations encountered two significant issues: a lost circulation event and a perforating gun misfire. These challenges impacted the efficiency and required corrective actions to ensure the well was completed to the required specifications.

2. Operations Overview

The completion of Well-3589-NX involved a series of operations to prepare and complete the wellbore for production. Following the drilling phase, casing was run and cemented in place. Perforating operations were conducted to connect the reservoir to the wellbore. The upper zone was successfully perforated and completed. However, the lower zone was isolated using a bridge plug at a specified depth. The completion was carried out in accordance with the planned sequence, though challenges were encountered during the process, necessitating additional steps to complete the well as intended.

3. Issues Encountered

During the completion of Well-3589-NX, a significant lost circulation event occurred at 1335 m MD, resulting in the loss of 200 m3 of drilling fluid. This event was addressed by applying two LCM pills to seal the lost circulation zone before proceeding with casing operations. The second issue encountered was a perforating gun misfire on run 3, where the guns across the interval 913-926 m MD did not fire. This required an additional perforating run to ensure the targeted interval was properly stimulated. Both issues added complexity to the completion process and required timely corrective actions to maintain the integrity of the well and ensure the completion met the required objectives.

4. Recommendations

To prevent similar issues in future operations, it is recommended to pre-treat the mud system with lost-circulation material before entering known loss zones. Additionally, verifying detonator continuity at the surface and carrying a backup firing head for each perforating run will help mitigate the risk of misfires and ensure the reliability of perforating operations.